Time Series Analysis — 3

Agenda ⇒

- Stationarity →
  - Check Stationarity
    - ADF → Dickey fuller

(ARIMA family)

- MoBiplus ⇒

ACF / PACF → = Seasonality

S = 12, S = 7, S = 30

Recap →

Trend ✓

→ Seasonality ✓

= Residual ✓

- Stationary Ts → μ, σ^2, freq. amplitude
  - ↳ adf Test

Ho

Ha

MORiplus → ARIMA

Time series

(1) Recomposition

$$y(t) = b(t) + s(t) + e(t)$$

$$e(t) = y(t) - [b(t) + s(t)]$$

$e(t)$ → X

$[b(t) + s(t)]$ → $\hat{y}$

(2) Differencing / Differentiation

$$\frac{\Delta y}{\Delta t} = \frac{y_2 - y_1}{t_2 - t_1} = \frac{y_2 - y_1}{1} = y_2 - y_1$$

y.diff(1)

y.diff(2) ⇒ $y_{10} - y_8$, $y_5 - y_3$

ARIMA

Seasonality

S = 12

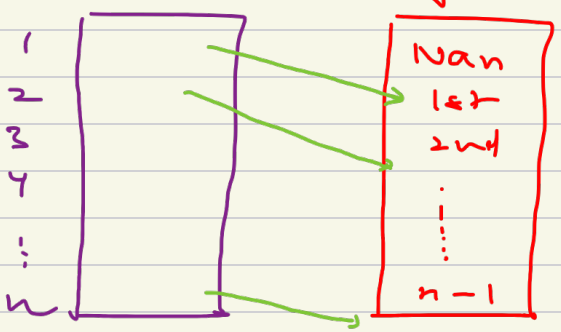
ACF | PACF →

MTS

original

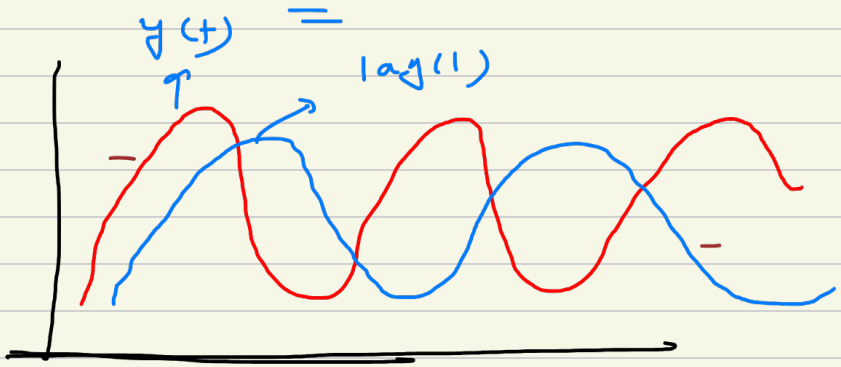
y(t)

lag(1) →



D1

D2



ok

PCC — Pearson correlⁿ coeff.

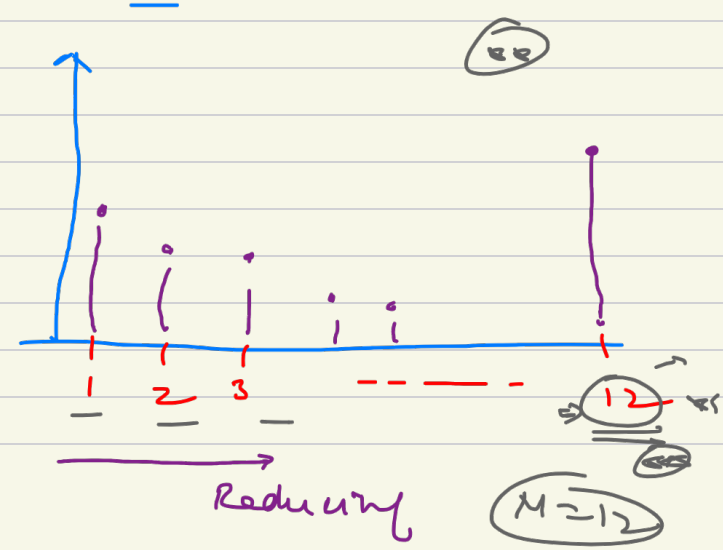
$y(t)$ & $y(t)$ lag

Autocorrelation

lag (1)

$y(t)$

lag	correlⁿ
1	0.7
2 ← sums	0.6
3	⋮
⋮	
12	0.8



Seasonality

Repeats

Electricity consumption

Today → yesterday
→ 2 days
→ 3 day
→ 2 days

lag

1
2
⋮
7

0.87
0.70
0.55

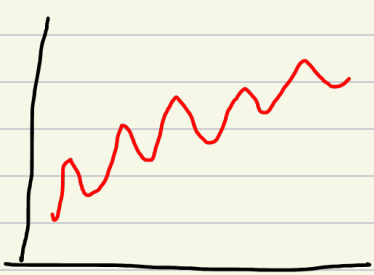
ACF → Seasonality

0.7 → ACF → NEXT

PACF → ⇒ ⇒

ARIMA →

→ TS



Trend, seasonality,
Residual

$$y(t) = b(t) + s(t) + e(t)$$

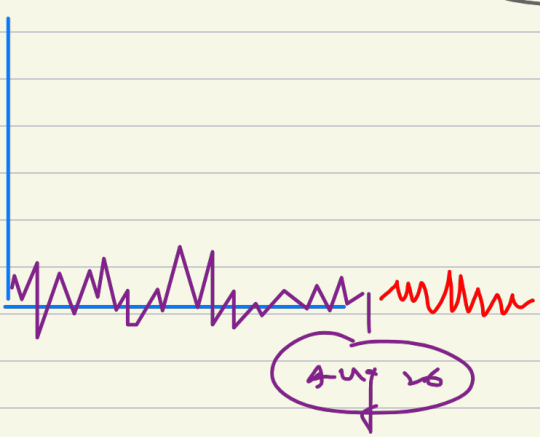
Still some predictable component

$$y(t) = b(t) + s(t) + e(t)$$

$e(t)$ → $z(t)$ + $e'(t)$

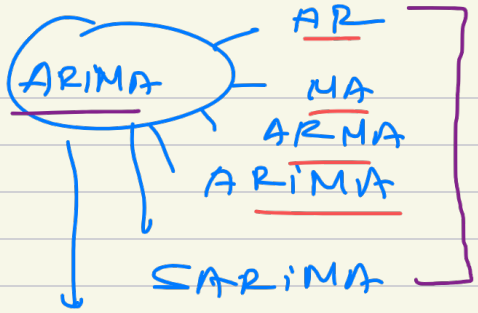
$e'(t)$: Truly unpredictable

ARIMA

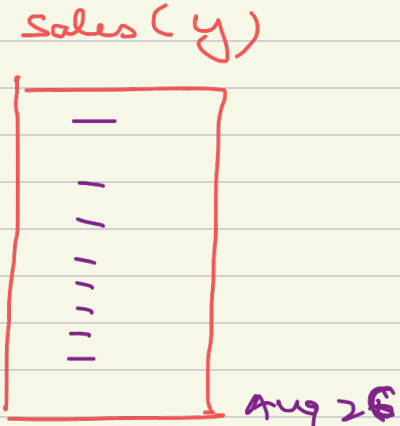


$e(t)$ → predictable

$e'(t)$ → Noise



SARIMAX → use case

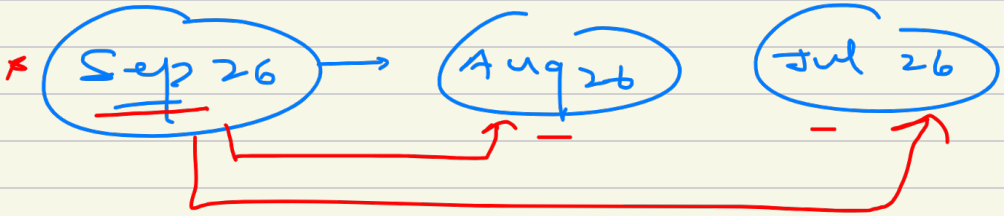


Aug 26

Next

Hyperparam

AR (p) → Auto Regressive



$$\hat{y}_t = \alpha_1 y_{t-1} + \alpha_2 y_{t-2} + \alpha_0$$

$p = 2$

2 months?

$$\hat{y}_t = \alpha_1 y_{t-1} + \alpha_2 y_{t-2} + \alpha_3 y_{t-3} + \alpha_0$$

$p = 3$

LR

AR(2) → wks. → ✓ model

X

X hyperparam

α → X

p

TES
DES
SES

generic

AR(p) ex

$$\hat{y}_t = \alpha_1 y_{t-1} + \alpha_2 y_{t-2} + \cdots \alpha_p y_{t-p} + \alpha_0$$



Auto Regressive

self data

α_1 α_2 α_p → learn these wts?

Linear Regn → α_s

$$\hat{y}_t = \underbrace{\alpha_1}_{w_1} \underbrace{y_{t-1}}_{f_1} + \underbrace{\alpha_2}_{w_2} \underbrace{y_{t-2}}_{f_2} + \cdots + \underbrace{\alpha_p}_{w_p} \underbrace{y_{t-p}}_{f_p} + \underbrace{\alpha_0}_{w_0}$$

Sales → supervised

AR(3)

27
14
15
20
25
30
37
26
19

ARIMA

NS → Stationary

Supervised

y_{t-3}	y_{t-2}	y_{t-1}	y_t
27	14	15	20
14	15	20	25
15	20	25	30

AR(2) ✗

α α α

AR(3) → LR.fit(X, y)

α_1 α_2 α_3 α_0

27
14
15
20
25
30
37
26
19

Jun 26
July
Aug 26

Sep 26

$\alpha_1 \cdot 19 + \alpha_2 \cdot 26 + \alpha_3 \cdot 87 + \alpha_0$

↓ pred^n = 20

α → importance

implementation

Moving avgs → MA(q) → Hyper param

forecasting

ARIMA →

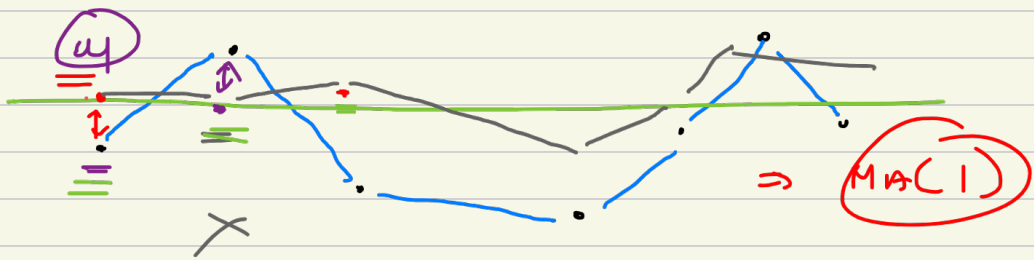
MA

today · unexpected errors

(q time periods)

daily food delivery orders

Day	What happened	Orders
Monday	Normal day	100
Tuesday	Unexpected heavy rain	140
Wednesday	Normal conditions	120
Thursday	Normal conditions	105
Friday	Normal again	100



MA

last month pred^n = avg

Overforecasted

→ underforecast →

MA(1)

LPP

MA(1)

MA → prev. mi stakes

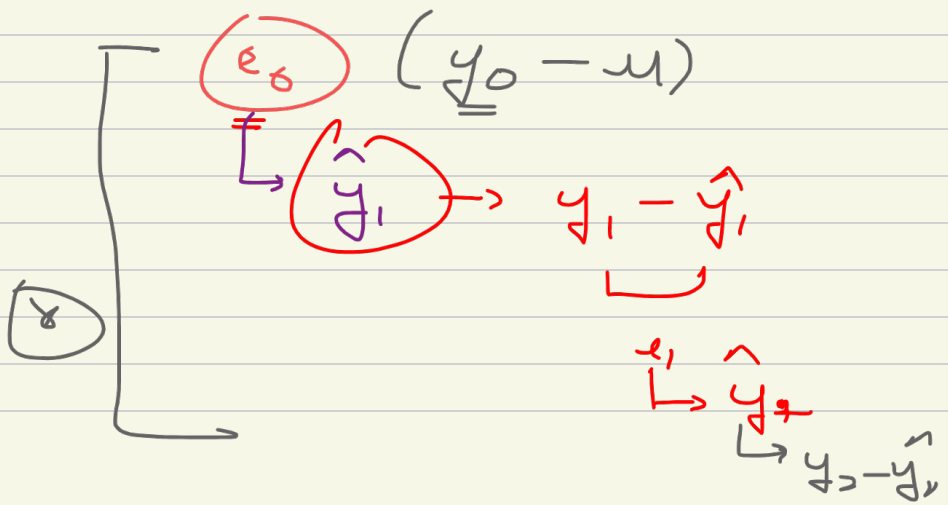
$\hat{y}_t = \mu + \beta_1 e_{t-1}$ MA(1)

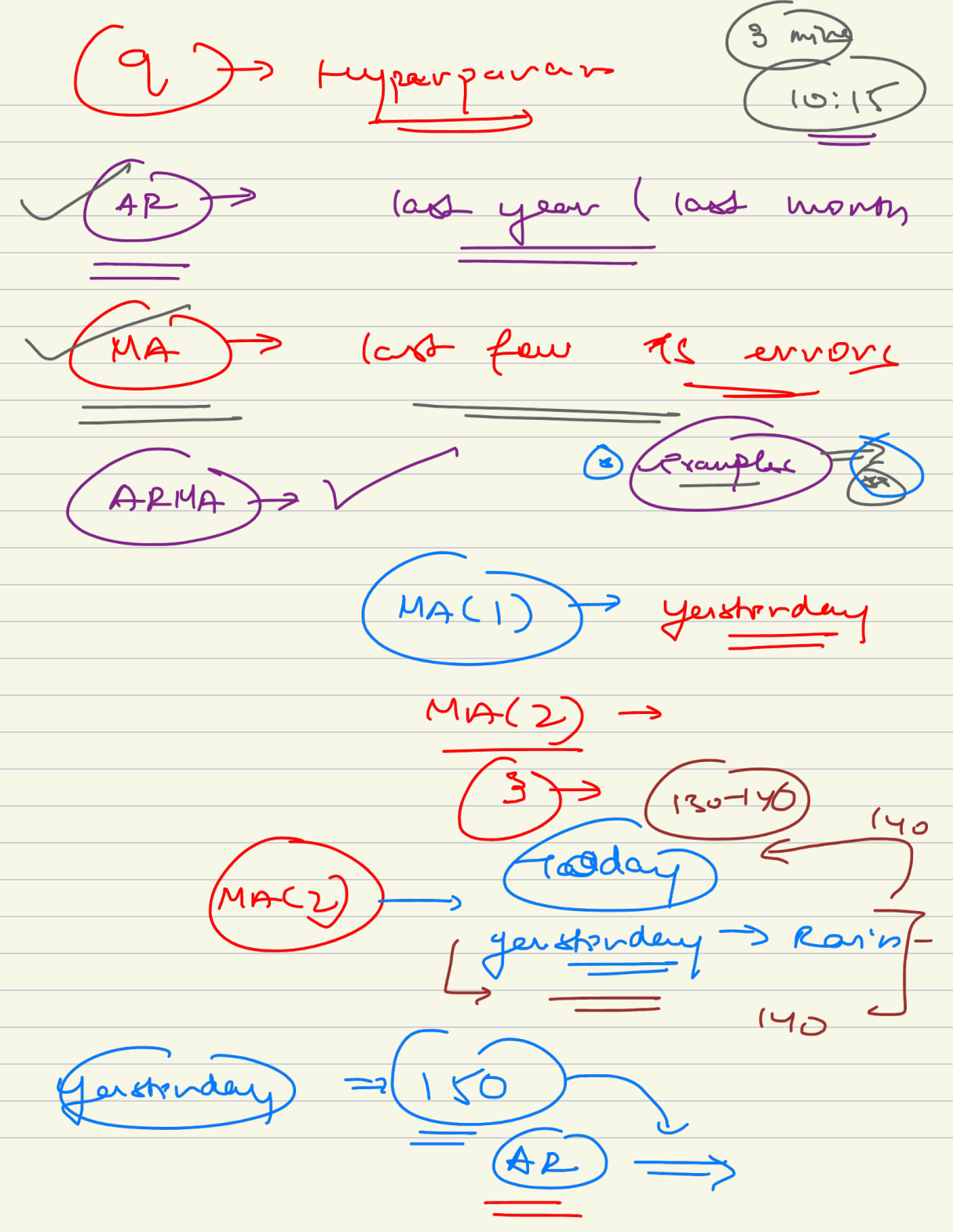
how much error y made at $t-1$

MA(3)

$y_t = \mu + \beta_1 e_{t-1} + \beta_2 e_{t-2} + \beta_3 e_{t-3}$ → global avg

MA(q)




q → Hyperparam
3 mins
10:15
AR → last year / last month
MA → last few TS errors
ARMA → ✓
Examples
MA(1) → yesterday
MA(2) →
3 → 130-140
140
today
MA(2) →
yesterday → Rain
140
Yesterday ⇒ 150
AR

MA

150

= diff

100

& 150

Expec = 100

Actual = 150

ERROR = 50

↓

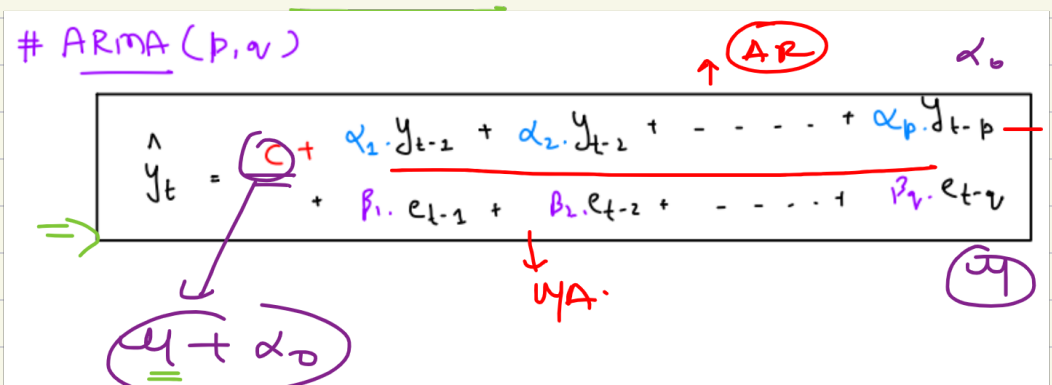
Surprise

MA

ARMA

AR + MA

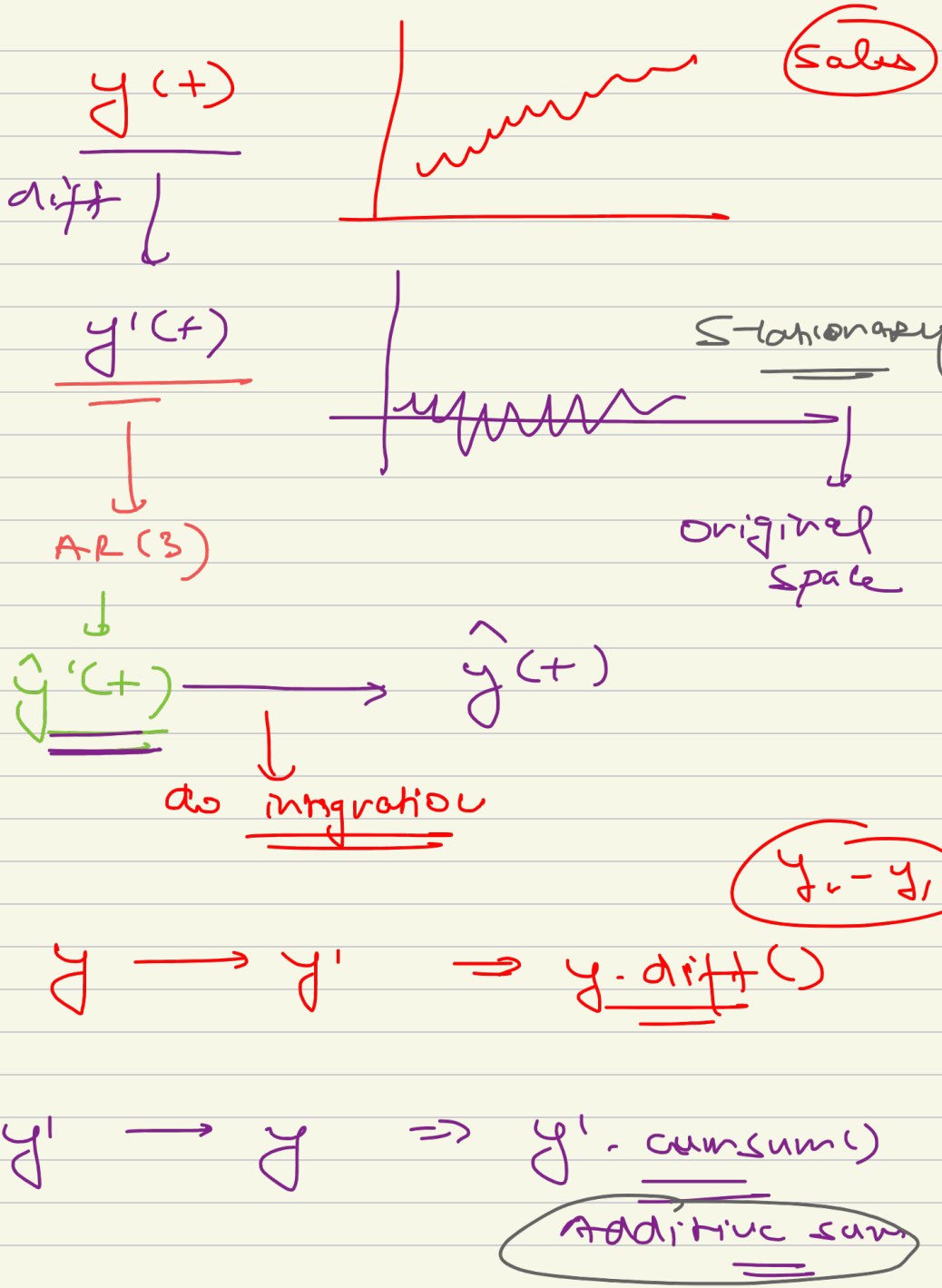
(p, q)



Stationary

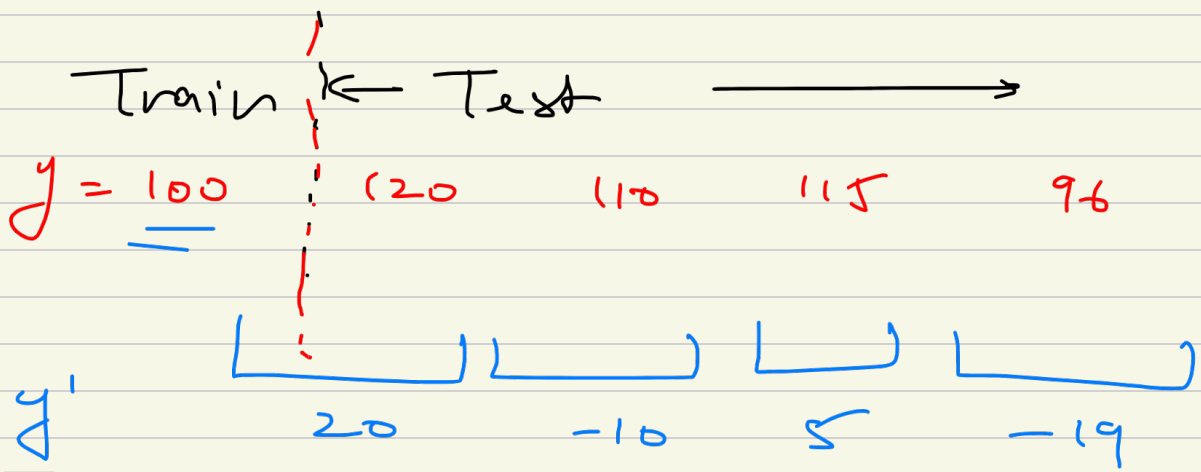
actual

(P) Post Processing


y(t)
diff
Sales
y'(t)
Stationary
AR(3)
original space
ŷ'(t)
ŷ(t)
do integration
y₁ - y₀
y → y' ⇒ y.diff()
y' → y ⇒ y'.cumsum()
Additive sum

1, 3, 2, 5, 19

cum sum → (1, 4, 6, 11, 15)



stationary ✓

AR / MA / ARMA → predn

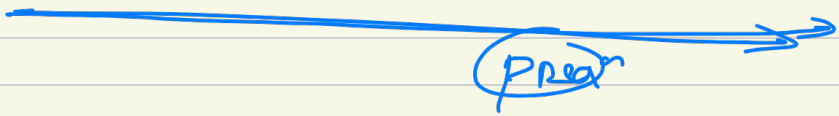
$\hat{y}'$ = 19 -7 +3 -17

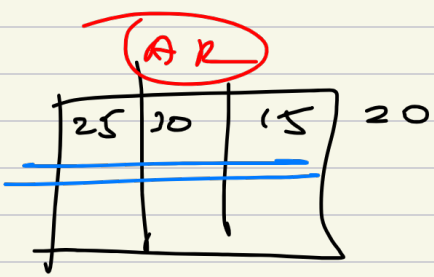
(predn)

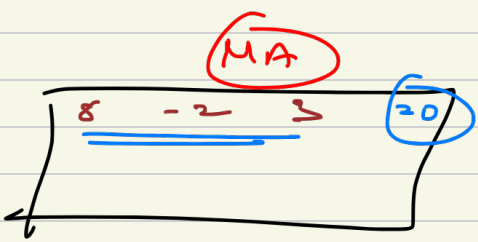
↓ cum sum

$\hat{y} \Rightarrow 100 + 19 \rightarrow 119 + (-7) \rightarrow 115$

$119 \qquad = 112 \rightarrow 98$





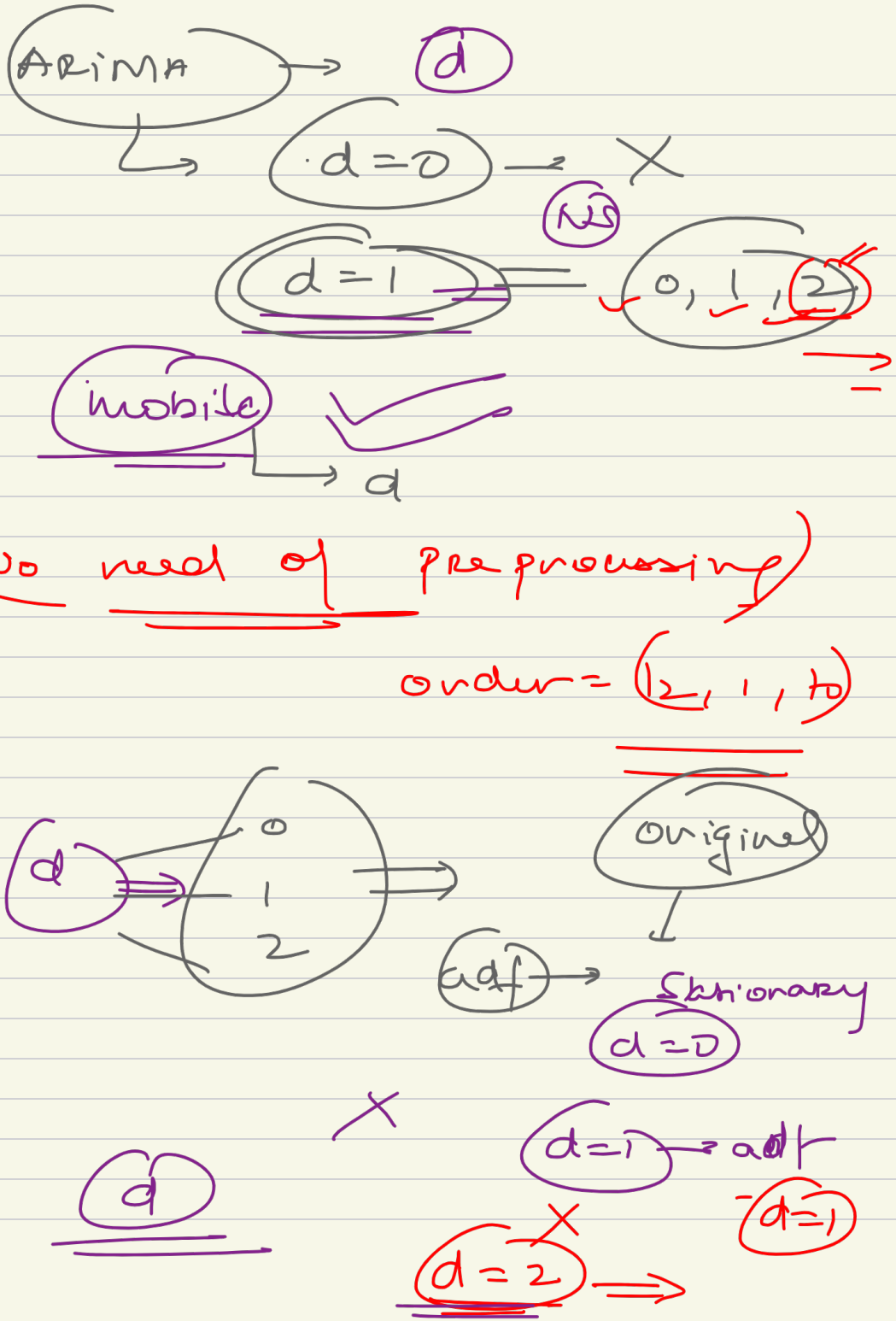


ARIMA → (p, d, q)

ARIMA → integrate

d → 0, 1, 2

pipeline ⇒ original → diff → AR / MA / ARMA → integrate → pred"


ARIMA
d
d=0
X
NS
d=1
0, 1, 2
mobile
d
(No need of preprocessing)
order = (2, 1, 10)
d
0
1
2
original
adf
Stationary
d=0
X
d=1
adf
d=1
d
d=2

ARIMA

order

p, d, q

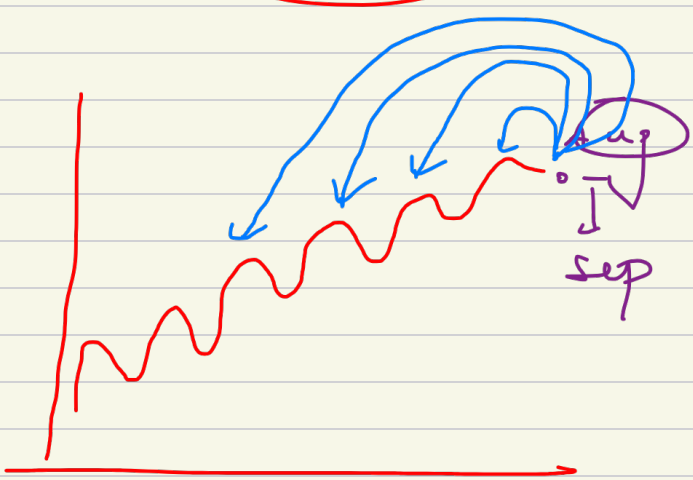
ARMA

MA , AR

Seasonality

SARIMA (p, d, q, P, D, Q, S)

ARIMA + Seasonal component



S

SARIMA

Seasonal component

AR → (last, 2nd last, 3rd last ---- 12)

$(y_{t-1} \quad - - - - \quad y_{t-12})$

MA

last month error,
2nd last error,
⋮
10th last month error

e_{t-1} e_{t-2} --- e_{t-q}

AR(p) =

$$\hat{y}_t = \alpha_1 y_{t-1} + \alpha_2 y_{t-2} + \cdots \alpha_p y_{t-p} + \alpha_0$$

SAR(P) S = 12 ACF

$$\hat{y}_t = \gamma_1 y_{t-1.s} + \gamma_2 y_{t-2.s} + \cdots \gamma_p y_{t-p.s} + \gamma_0$$

$\hat{y}_t$: Sep 26
$y_{t-1.s}$: Sep 25
$y_{t-2.s}$: Sep 24

LR

MA(q)

SMA(Q) ⇒

last yr

sep

sep LY

sep 22

$$\hat{y}_t = \delta_1 e_{t-1.s} + \delta_2 . e_{t-2.s} + \cdots \delta_q e_{t-q.s} + \delta_0$$

Ex

SARIMA (p, d, q, P, D, Q, S)

SARIMA (p, d, q, P, D, Q, s)

$$\hat{y}_t = C + \alpha_1 . y_{t-1} + \alpha_2 . y_{t-2} + \cdots + \alpha_p . y_{t-p} \rightarrow AR$$

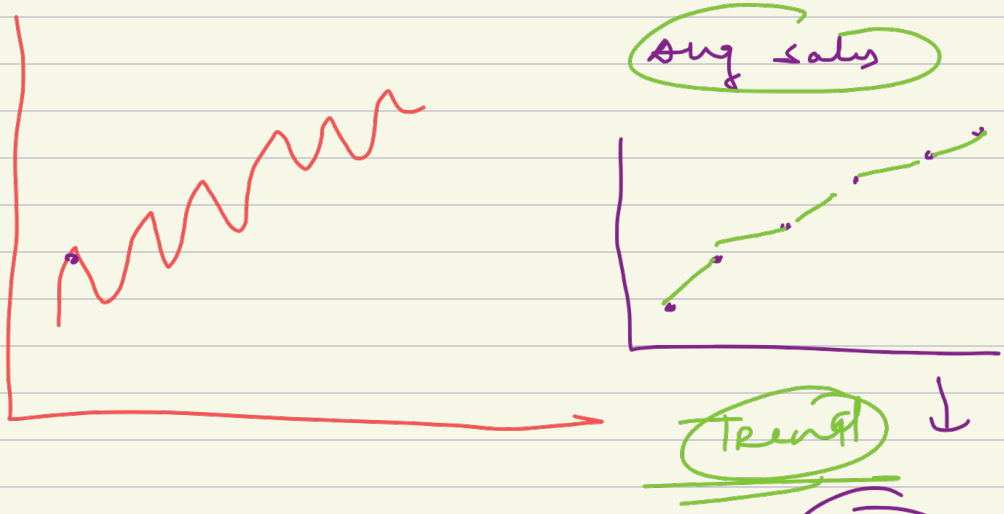
$$+ \beta_1 e_{t-1} + \beta_2 e_{t-2} + \cdots + \beta_q e_{t-q} \rightarrow MA$$

$$+ \gamma_1 . y_{t-1s} + \gamma_2 . y_{t-2s} + \cdots + \gamma_P . y_{t-Ps} \rightarrow SAR$$

$$+ \delta_1 . e_{t-1s} + \delta_2 . e_{t-2s} + \cdots + \delta_Q e_{t-Qs} \rightarrow SMA$$

idea

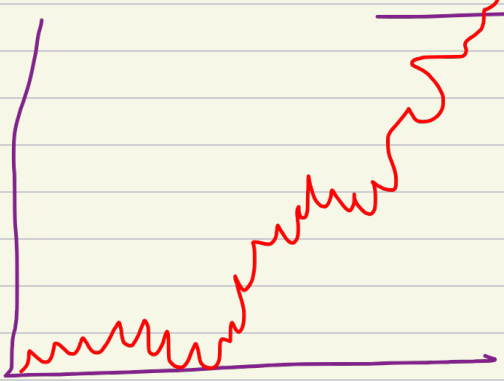
D



NS

D → Stationary

D=2



⇒ Trend ↑ linearly

NO

$b(t) = m_1 t^2 + m_2 t + c$

d=2 $b'(t) = 2m_1 t + m_2$

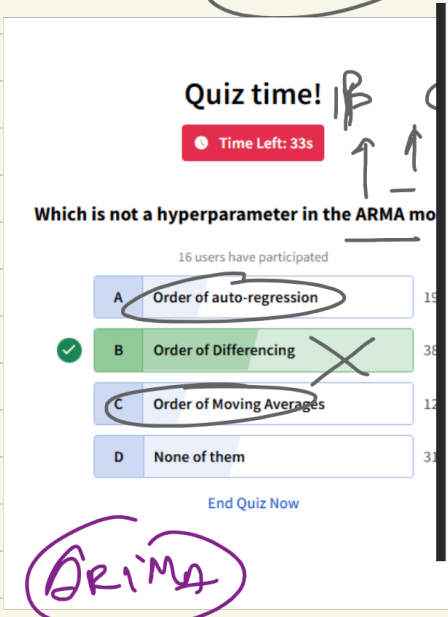
D=2 ↓

$b''(t) = 2m_1$ → No Time component ⇒

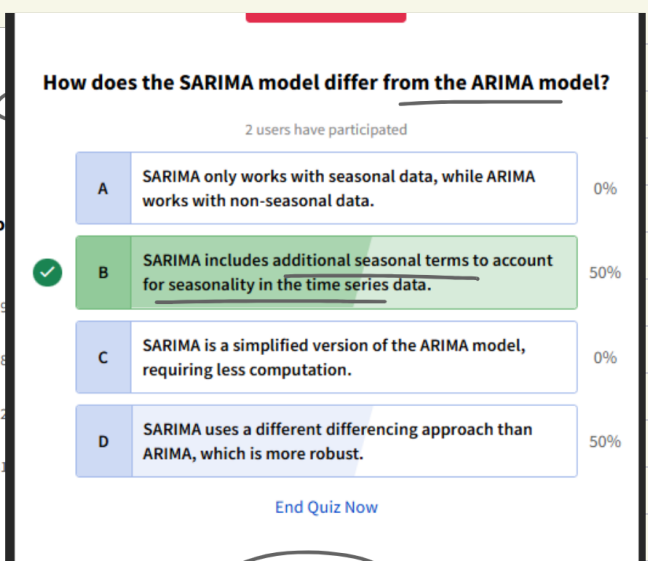
D ⇒ seasonality

d = ⇒ ✓

D → seasonality



ARIMA



DOUBT